

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	Individual Project
Project Name	Book - A - Doctor
Maximum Marks	4 Marks

Functional Requirements

FR-1: User Registration

Sub Requirements:

- Patient Registration through Registration Form
- Doctor Registration through Registration Form
- Validation of User Details
- Secure Password Storage

FR-2: User Authentication

Sub Requirements:

- User Login using Email and Password
- JWT-Based Authentication
- Role-Based Access Control
- Logout Functionality

FR-3: Doctor Management

Sub Requirements:

- Doctor Profile Creation
- View Doctor Details
- Search Doctors by Specialization
- Update Doctor Information

FR-4: Appointment Booking

Sub Requirements:

- Book Appointment with Doctor
- Select Appointment Date and Time
- Appointment Confirmation
- Appointment Cancellation

FR-5: Appointment Management

Sub Requirements:

- View Appointment Requests
- Approve Appointment Requests
- Reject Appointment Requests
- Track Appointment Status

FR-6: Medical Report Management

Sub Requirements:

- Upload Medical Reports
- View Uploaded Reports
- Store Reports Securely
- Associate Reports with Patient Profile

FR-7: Dashboard Management

Sub Requirements:

- Patient Dashboard
- Doctor Dashboard
- Appointment History
- Profile Information Display

FR-8: Database Management

Sub Requirements:

- Store User Information

- Store Doctor Information
- Store Appointment Details
- Store Medical Reports

Non-Functional Requirements

NFR-1: Usability

Description:

The application should provide a simple, intuitive, and user-friendly interface that allows patients and doctors to navigate the system easily without extensive training.

NFR-2: Security

Description:

The system should implement JWT authentication, password encryption, secure API communication, and role-based access control to protect user data and prevent unauthorized access.

NFR-3: Reliability

Description:

The application should consistently perform healthcare appointment operations without data loss, system crashes, or unexpected failures.

NFR-4: Performance

Description:

The system should process user requests, load pages, and retrieve appointment information quickly with minimal response time under normal operating conditions.

NFR-5: Availability

Description:

The application should remain accessible to users whenever required and support continuous operation with minimal downtime.

NFR-6: Scalability

Description:

The system should support an increasing number of patients, doctors, appointments, and medical records without significant degradation in performance.

NFR-7: Maintainability

Description:

The application should be designed using modular architecture and coding standards so that future updates, bug fixes, and feature enhancements can be implemented efficiently.

NFR-8: Compatibility

Description:

The system should function properly across different web browsers and devices, including desktops, laptops, tablets, and smartphones.

NFR-9: Data Integrity

Description:

The application should ensure accuracy, consistency, and reliability of patient, doctor, appointment, and medical report data stored in the database.

NFR-10: Responsiveness

Description:

The user interface should adapt effectively to different screen sizes and provide a consistent experience across multiple devices.